CHAPTER – 1

INTRODUCTION

A power transformer transfers electrical energy between circuits through electromagnetic induction and is an essential component of electrical generation, transmission, and distribution systems. Its reliability depends on operating conditions such as voltage, current, loading, temperature, insulation condition, and mechanical behaviour. Conventional protection systems respond to defined electrical faults, while condition monitoring helps identify abnormal operating conditions through continuous measurement and condition assessment.

A smart transformer monitoring system combines electrical measurements, condition indicators, diagnostic logic, and a monitoring interface to assess operating conditions. This project develops a simulation-based transformer health monitoring system using MATLAB/Simulink. The model is evaluated under normal and selected abnormal operating conditions to study changes in electrical parameters and demonstrate condition assessment before proceeding towards a physical prototype.

1.1 Problem Statement :

Transformer operating conditions can change with variations in supply voltage, loading, and fault conditions. A monitoring system must identify these changes and present useful information for condition assessment. Developing and testing the monitoring logic directly on hardware can make troubleshooting difficult and costly.

Therefore, a simulation-based approach is adopted to study transformer behaviour under repeatable operating conditions, verify calculated electrical parameters, and evaluate the implemented monitoring and diagnostic functions before hardware development.

1.2 Aim :

To develop and evaluate a three-phase transformer health monitoring model using MATLAB/Simulink that monitors electrical parameters, generates simulated temperature and vibration indicators, identifies selected abnormal operating conditions, and presents the results through a monitoring dashboard.

1.3 Objectives :

The main objectives of the project are:

1. To model a **5 kVA, 400/230 V, three-phase transformer** with delta-connected primary and grounded-star secondary windings.
2. To monitor secondary phase RMS voltages, phase RMS currents, loading percentage, and apparent-power estimates.
3. To simulate normal operation, 150% and 200% overload target conditions, overvoltage, undervoltage, and unbalanced loading.
4. To examine open-phase and short-circuit responses where implemented in the simulation.
5. To observe simulated temperature and vibration indicators and evaluate the displayed condition score, system status, and available alarm flags.
6. To compare calculated references with displayed simulation results and document the findings and limitations.

1.4 Scope and Exclusions :

In scope: MATLAB/Simulink simulation, Simscape Electrical power-path modelling, electrical measurement and signal conditioning, fault-state logic, synthetic thermal/vibration indicators, plots and a validation report. Out of scope for the present semester: building the physical sensor board, ESP32 firmware, live IoT connection, certified protection, dissolved-gas analysis, insulation diagnosis, exact remaining useful life prediction and validation against an actual transformer.

CHAPTER – 2

LITERATURE SURVEY

The literature survey covers transformer standards, MATLAB/Simulink documentation, and published research on transformer condition monitoring.

2.1 Standards and Technical Documentation :

- **IEC 60076-1 – Power Transformers [1]:** Defines general transformer requirements, ratings, and technical characteristics.
- **MathWorks Transformer Model [2]:** Supports three-phase transformer simulation with configurable winding connections and electrical parameters.
- **MathWorks Sensor Documentation [3], [4]:** Specifies correct voltage and current sensor connections for electrical measurements.

2.2 Research and Industrial Approaches :

- **CIGRE Condition Monitoring [5]:** Highlights sensor integration, continuous monitoring, and data analysis for transformer condition assessment.
- **IEEE Thermal Monitoring Review (2022) [6]:** Discusses transformer heating, temperature distribution, and hotspot monitoring.
- **Transformer Health Index Review (2025) [7]:** Examines conventional and machine-learning methods for health assessment and lifespan prediction.

2.3 Summary of Literature :

Reference	Main Contribution	Application in This Project
IEC 60076-1	Defines transformer ratings and requirements.	Establishes transformer specifications.
MathWorks Transformer Model	Supports three-phase transformer simulation.	Models the 5 kVA transformer.
MathWorks Sensor Documentation	Explains sensor connections.	Guides voltage and current measurement.
CIGRE Monitoring Guide	Presents integrated condition monitoring.	Guides system architecture.
MathWorks Sensor Documentation	Explains sensor connections.	Guides voltage and current measurement.
IEEE Thermal Monitoring Review (2022)	Explains transformer thermal behaviour.	Supports temperature monitoring.
Health Index Review (2025)	Reviews transformer health assessment methods.	Informs health indication

Table 2.1: Summary of Literature Survey

CHAPTER – 3

PROJECT PLANNING AND WORK

3.1 Design Philosophy :

The project follows a systematic approach: develop the transformer model in MATLAB/Simulink, measure voltage and current under different operating conditions, analyse the simulation results, and study changes in transformer health.

Stages	Activities	Timeline
Model Development	Develop and run the transformer model in MATLAB/Simulink.	Weeks 1–2
Measurement Setup	Obtain voltage and current readings.	Week 3
Simulation and Testing	Test normal, varying-load, and selected fault conditions.	Weeks 4–5
Result Analysis	Analyse simulation graphs and record observations.	Week 6
Report Preparation	Compile the findings and complete the project report.	Week 7

Table 3.1: Project Work Plan and Timeline

Item	Description
Software	MATLAB/Simulink
Model	Transformer simulation model
Measurements	Voltage and current readings
Results	Simulation graphs and observations
Documentation	Final project report

Table 3.2: Tools and Deliverables

Note: *The timeline is tentative and may be revised according to project progress. Additional features may be incorporated during further development.*

CHAPTER – 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture :

The implemented system is a MATLAB/Simulink simulation organised into three linked sections: the three-phase electrical plant, measurement and RMS signal processing, and the Health Monitor subsystem. The electrical section models the supply, transformer, switching/fault elements and load. Measured voltage and current signals are processed phase-by-phase and supplied to the monitoring subsystem, which produces condition values and diagnostic flags.

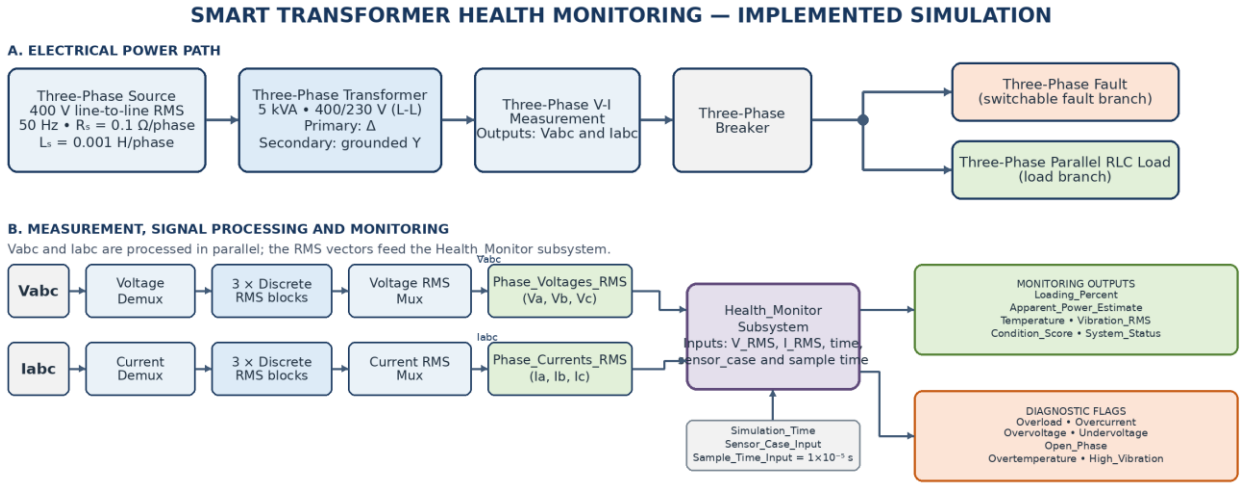


Figure 4.1: Functional architecture and signal flow of the implemented MATLAB/Simulink model

4.2 Electrical Plant Model :

The power circuit uses a three-phase source, a two-winding transformer, a Three-Phase V-I Measurement block, a three-phase breaker, a Three-Phase Fault block and a Three-Phase Parallel RLC Load. The model uses a 5 kVA, 400/230 V line-to-line transformer with a delta primary and grounded-star secondary. The confirmed configuration values are summarised in Table 4.1.

Parameter	Confirmed setting	Clarification
Three-phase source	400 V line-to-line RMS; 50 Hz	Input source setting
Source internal impedance	R_s = 0.1 Ω/phase; L_s = 0.001 H/phase	Values shown in source mask
Transformer	5 kVA; 400/230 V line-to-line; Δ / grounded Y	Nominal model description

Parameter	Confirmed setting	Clarification
Load and switching	Parallel RLC Load; three-phase breaker and fault block	-
Simulation configuration	Stop time = 2 s; discrete sample time = 1×10^{-5} s	-

Table 4.1: Confirmed electrical model and simulation settings

4.3 Measurement and Signal Processing :

The Three-Phase V-I Measurement block supplies the voltage vector (Vabc) and current vector (I_{abc}). Each vector is separated into phase components by a Demux block. Three Discrete RMS blocks calculate the RMS value of each phase, and a Mux block combines the phase results into Phase Voltages RMS and Phase Currents RMS. These vectors are connected to the Health Monitor inputs V_{RMS} and I_{RMS}; voltage and current scopes display the corresponding waveforms.

Signal path	Processing blocks	Result / purpose
Vabc	Voltage Demux → 3 × Discrete RMS → Voltage RMS Mux	Phase Voltages RMS: V_a, V_b and V_c RMS values
Iabc	Current Demux → 3 × Discrete RMS → Current RMS Mux	Phase Currents RMS: I_a, I_b and I_c RMS values
Supporting inputs	Simulation Time; Sensor Case Input; Sample Time Input	Time/test-case inputs for Health Monitor; displayed sample time is 1×10^{-5} s
Output processing	Health Monitor subsystem and output displays	Loading, apparent-power estimate, temperature, vibration, condition score, status and flags

Table 4.2: Measurement and monitoring signal paths

RMS and Loading Calculations

1. RMS Voltage and Current

$$V_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} v_k^2}$$

$$I_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} i_k^2}$$

2. Rated Secondary Line Current

$$I_{2,\mathrm{rated}} = \frac{S_{\mathrm{rated}}}{\sqrt{3}\,V_{2,LL}} = \frac{5000}{\sqrt{3}\times 230} = \boxed{12.5511\ \mathrm{A}}$$

3. Loading Percentage

$$\text{Loading (\%)} = \frac{\max\left(I_{a,\text{RMS}}, I_{b,\text{RMS}}, I_{c,\text{RMS}}\right)}{12.5511} \times 100$$

4. Apparent-Power Estimate

$$S_{est} = 3V_{ph,RMS} \times I_{ph,RMS}$$

For a balanced three-phase system:

Using the maximum phase RMS current is consistent with the model's unbalanced-load display; for balanced cases, all three phase currents are equal and this reduces to the current-based cross-check in Appendix A.8. The apparent-power expression is an estimate calculated from the displayed phase RMS values, as cross-checked in Appendix A.7; it is not active power.

4.4 Condition Monitoring and Diagnostic Outputs :

The Health Monitor subsystem receives phase RMS voltage and current values together with simulation time, the sensor/test-case input and the sample-time input. Its displayed outputs are Loading Percent, Apparent Power Estimate, Temperature, Vibration RMS, Condition Score, System Status and the diagnostic flags listed below.

Output group	Signals shown in the model
Condition indicators	Loading Percent; Apparent Power Estimate; Temperature; Vibration RMS; Condition Score; System Status
Diagnostic flags	Overload; Overcurrent; Overvoltage; Undervoltage; Open Phase; Overtemperature; High Vibration

Table 4.3: Health monitoring outputs available in the model

4.4.1 Interpretation of Monitoring Outputs and System Status :

The Health Monitor subsystem generates condition indicators and diagnostic outputs to assess the transformer's operating condition.

Output	Description
Loading (%)	Maximum phase RMS current relative to the rated secondary current.
S_{est}	Estimated apparent power in VA.
T	Modelled temperature indicator in °C.
V_{ib}	Synthetic vibration RMS indicator in m/s².
HI	Project-specific condition score ranging from 0 to 100.

Output	Description
Status	Overall system condition code.
Flags	Seven individual fault indicators.

Table 4.4: Description of Health Monitoring Outputs

The overall system status is interpreted as follows:

Status Code	Condition
0	Normal
1	Warning
2	Fault

Table 4.5: System Status Code Interpretation

The seven diagnostic flags represent **overload, overcurrent, overvoltage, undervoltage, open phase, overtemperature, and high vibration**. These flags indicate the corresponding conditions when activated.

Note: *The temperature and vibration outputs are model-generated indicators rather than physical sensor measurements.*

4.5 Simulation Procedure :

1. Configure the three-phase source, transformer winding connections, measurement blocks, breaker, load and Health Monitor inputs using the available model settings.
2. Run the baseline case at 400 V line-to-line and 50 Hz for the 2 s simulation interval; record the three phase RMS voltages and currents, dashboard outputs and waveforms.
3. Change one operating condition at a time (load level, source voltage, phase loading or fault switch) and repeat the simulation, retaining the parameter values used for each case.
4. Compare measured RMS values and dashboard outputs with the theoretical references in Appendix A, including the rated secondary current, phase-voltage reference and apparent-power estimate.
5. Record an alarm or fault-detection result only when the corresponding output is visible and supported by the simulation evidence. Do not infer missing fault values or undocumented thresholds.

4.6 Test Conditions :

The model was assessed using the operating cases summarised in Table 4.4. Each case is considered separately so that the change in electrical quantities and the resulting monitoring outputs can be compared with the normal condition.

Test case	Input / model change	Purpose
Normal operation	Source = 400 V line-to-line, 50 Hz	Establish baseline RMS values and condition indicators

Test case	Input / model change	Purpose
150% overload target	Increase loading to the 150% target case	Observe current, loading and related flags
200% overload target	Increase loading to the 200% target case	Observe the response at a higher loading level
Overvoltage	Source = 440 V line-to-line	Examine the response to an increased supply voltage
Undervoltage	Source = 360 V line-to-line	Examine the response to a reduced supply voltage
Unbalanced load	Unequal phase loading	Compare phase currents and voltages
Open phase	Phase A path opens after 0.8 s	Examine the response to an interrupted phase path
Short circuit	Fault event shown from 0.8 s to 1.0 s	Observe the waveform response during the fault interval

Table 4.6: Simulation test conditions

CHAPTER – 5

SIMULATION RESULTS AND DISCUSSION

5.1 Simulation Model :

The MATLAB/Simulink model combines a three-phase AC source, source impedance, a three-phase transformer, switching/fault elements, a three-phase load, RMS measurement blocks and a MATLAB Function subsystem named Health_Monitor. The monitor receives RMS voltage/current signals and simulation time and produces loading percentage, an apparent-power estimate, temperature and vibration indicators, a condition score, a status code and fault flags.

The source-parameter has 400 V line-to-line RMS, 50 Hz supply, source resistance of 0.1 Ω and source inductance of 1 mH. The report identifies the transformer as 5 kVA, 400/230 V, delta-connected primary and grounded-star secondary. The model uses a discrete powergui/sample time of 1×10^{-5} s and a simulation stop time of 2 s.

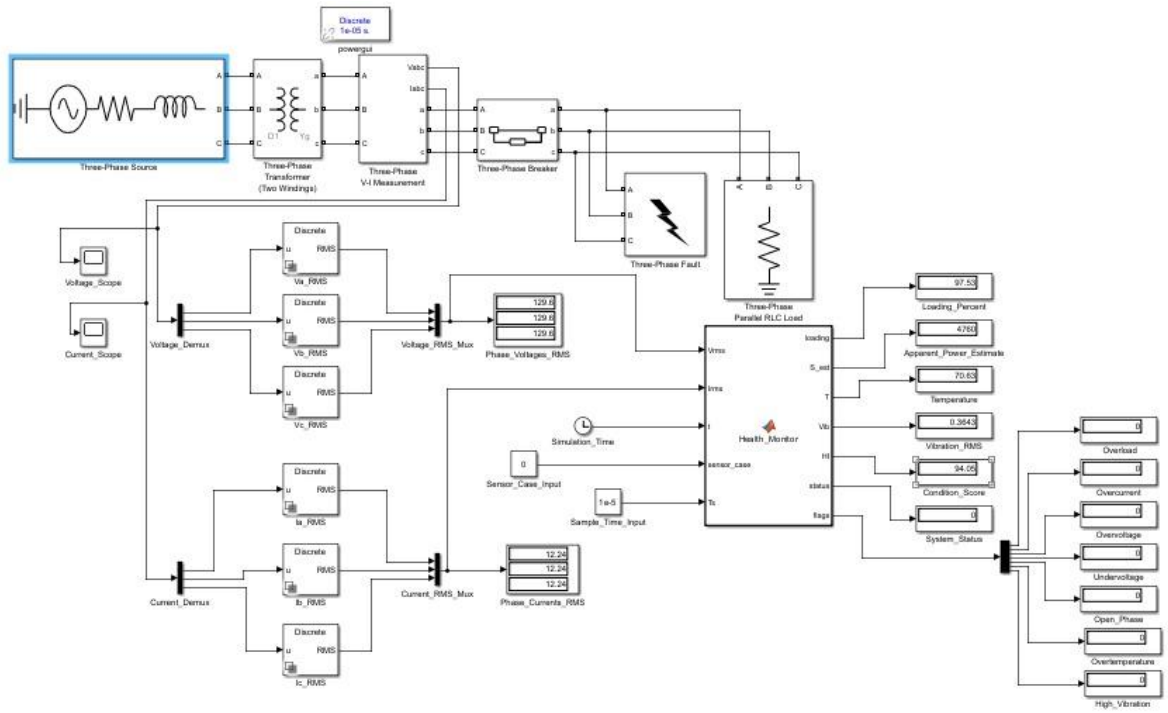


Figure 5.1: Complete MATLAB/Simulink transformer monitoring model

5.2 Normal Operating Condition :

The normal-case scope plots show three balanced, sinusoidal phase waveforms. The monitoring dashboard reports equal RMS values for all three phases, a loading estimate close to 100%, and all displayed fault flags at zero.

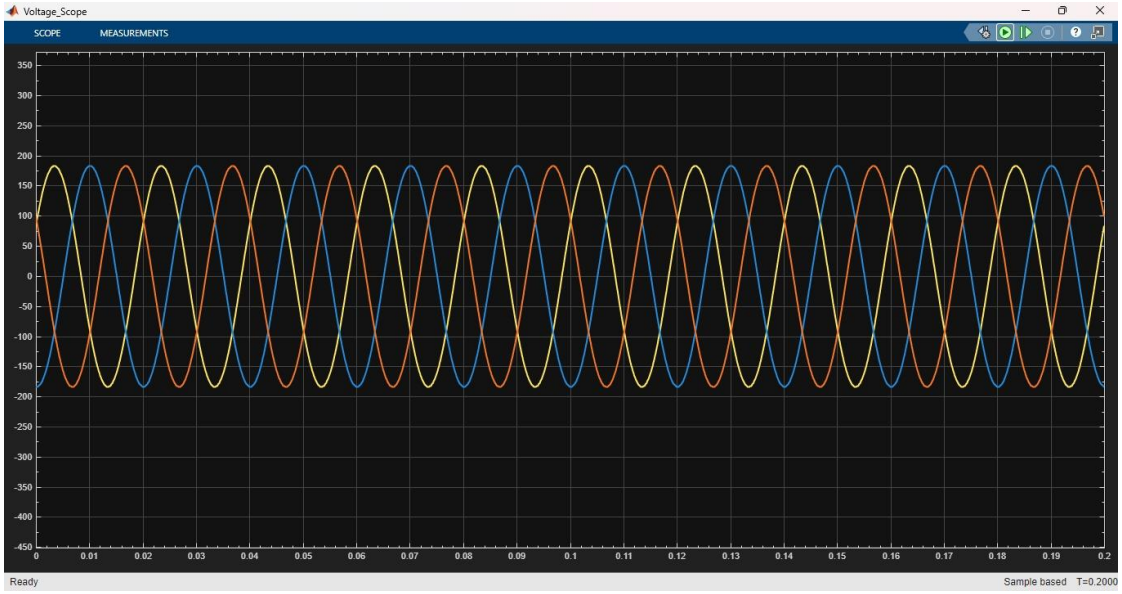


Figure 5.2(a): Three-phase secondary-voltage waveforms under normal operation

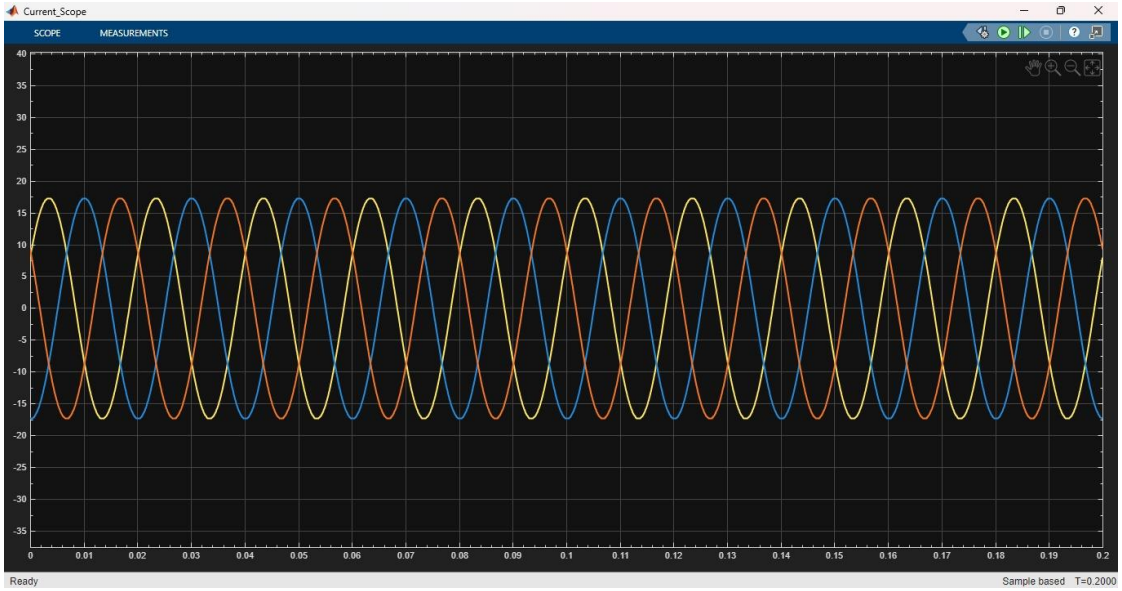


Figure 5.2(b): Three-phase current waveforms under normal operation

Parameter	Unit	Actual displayed/configured value	Reference / interpretation
Phase-voltage RMS (each phase)	V	129.6	132.79 V ideal phase-to-neutral
Phase-current RMS (each phase)	A	12.24	12.55 A rated line current
Loading Percent	%	97.53	Current-based monitor output
Apparent Power Estimate	VA	4,760	≈ 4.76 kVA
Temperature output	°C	46.29	-
Vibration RMS	displayed value	0.3643	-

Condition Score	score	94.05	-
Fault flags	binary	All 0	Overload, overcurrent, overvoltage, undervoltage, open phase, overtemperature, high vibration
Supply frequency	Hz	50 (configured)	-

Table 5.1: Normal operating measurements

The measured phase-voltage output of 129.6 V is about 2.40% below the ideal 132.79 V phase-to-neutral value. The measured current of 12.24 A is about 2.48% below the 12.55 A rated-current reference. The monitor's loading output of 97.53% is consistent with this current ratio. These differences are plausible in the implemented model but cannot be attributed to a particular transformer loss or impedance without the complete transformer and load parameters.

5.3 Measurement Parameters and Monitoring Display :

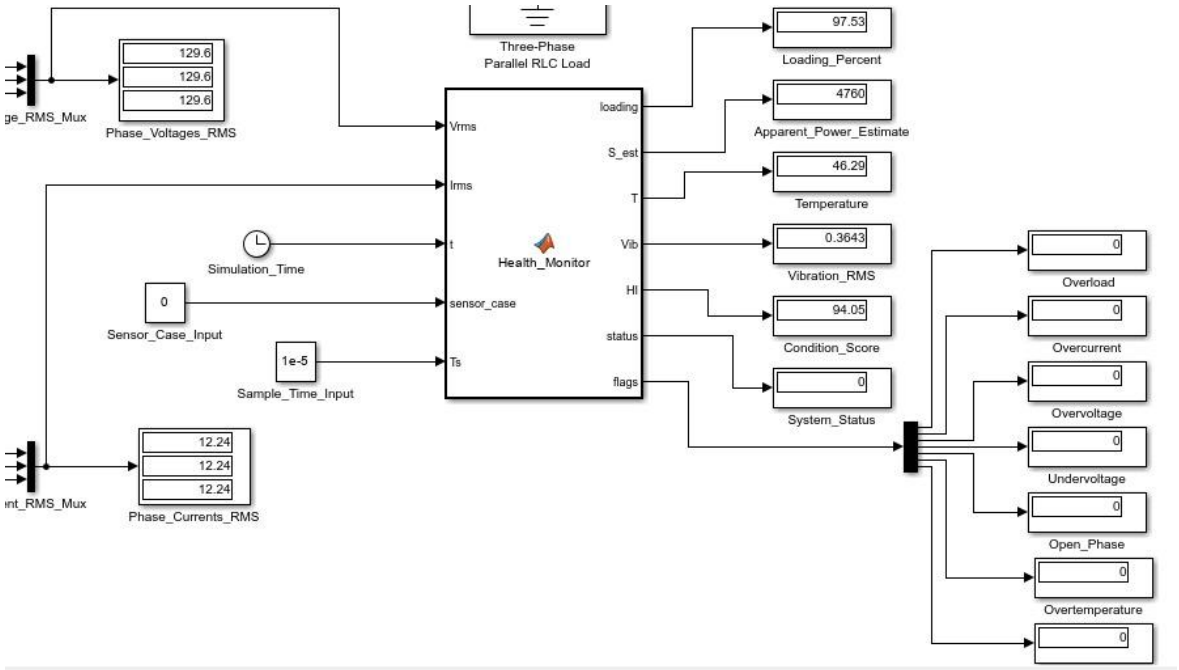


Figure 5.3: Health Monitor subsystem and its displayed outputs under normal operation

The model directly displays phase RMS voltage and current, loading percentage, apparent-power estimate, temperature, vibration RMS, condition score, system-status code and Boolean fault flags.

Output	**Normal**	**150% target case**	**200% target case**	**440 V source**	**360 V source**
Phase voltage RMS (per phase, V)	129.6	128.0	126.4	142.6	116.7

Phase current RMS (per phase, A)	12.24	18.12	23.84	13.46	11.02
Loading Percent (%)	97.53	144.4	189.9	107.3	87.77
Apparent power estimate (VA)	4,760	6,959	9,037	5,760	3,856
Temperature output (displayed)	46.29	123.2	191.2	81.43	64.43
Vibration RMS (displayed)	0.3643	0.4668	0.5886	0.3921	0.3616
Condition Score	94.05	47.46	24.1	74.69	75
System Status code	0	2	2	2	2

Table 5.2: Numerical values transcribed from the available monitoring-dashboard

5.4 Overload and Overcurrent Tests :

The two overload tests use lower per-phase resistance values to increase current. The 150% and 200% labels describe nominal target cases; the dashboard reports 144.4% and 189.9% loading based on its current-derived calculation.

5.4.1 Overload at 150% :

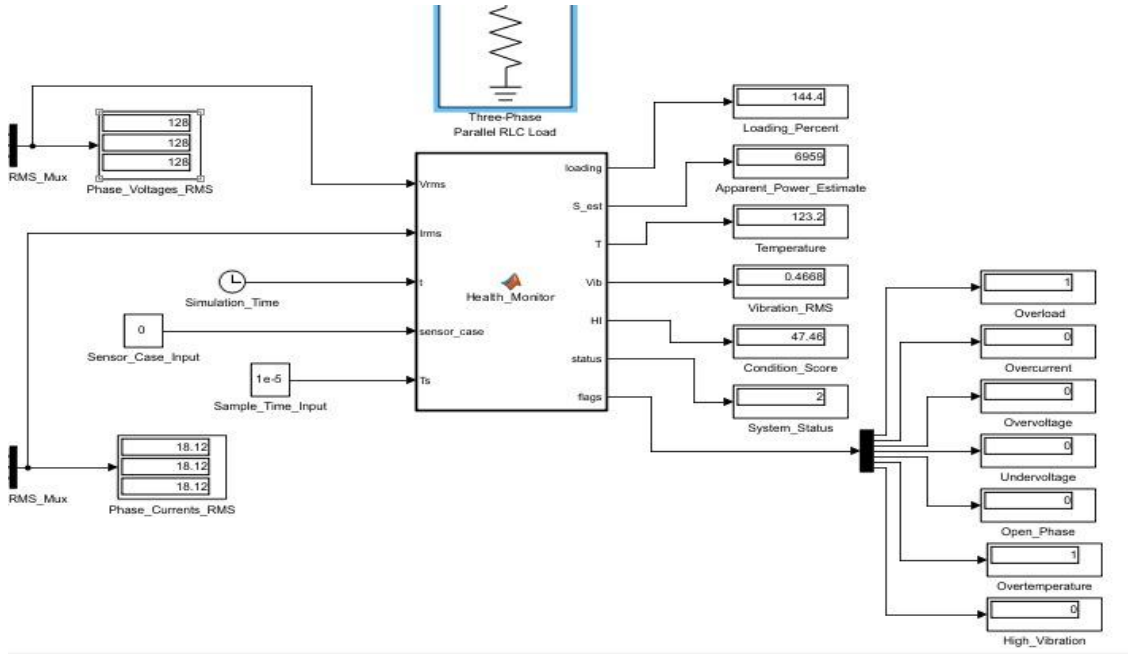


Figure 5.4.a: Simulation Response under 150% Loading

5.4.2 Overload at 200% :

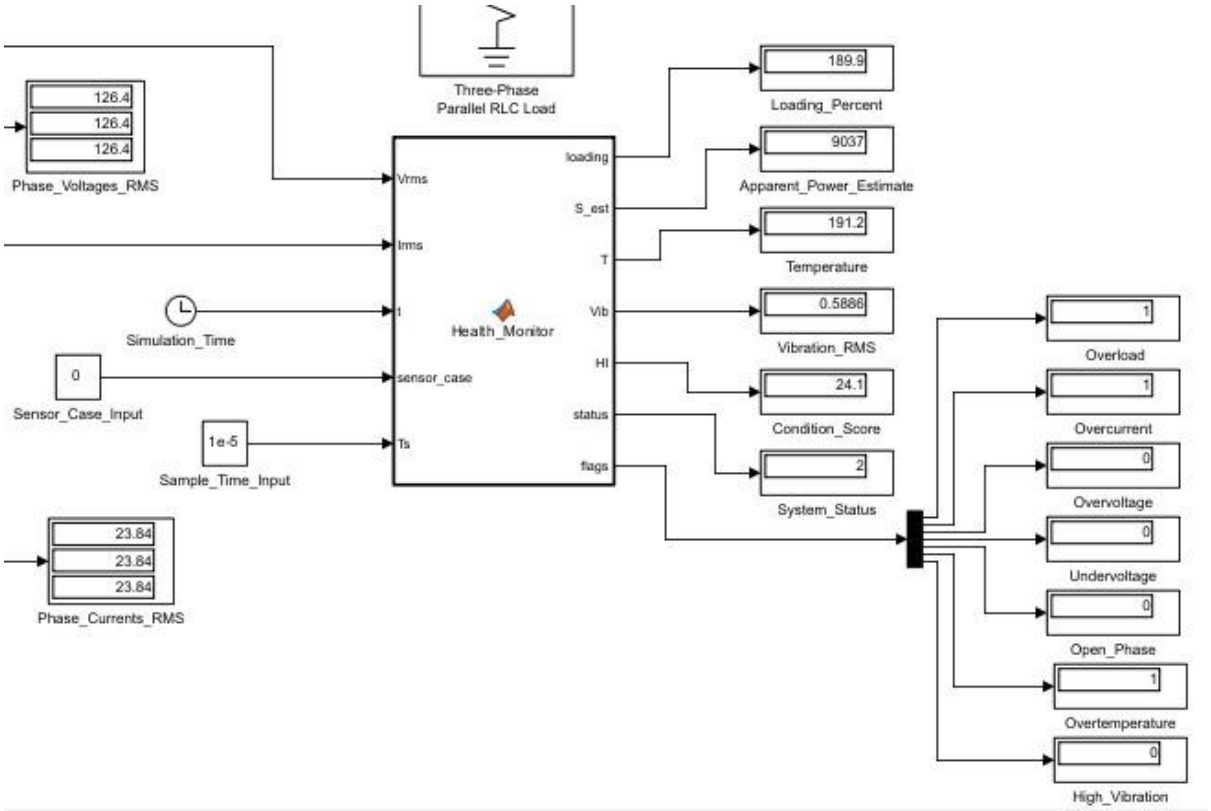


Figure 5.4.b: Simulation Response under 200% Loading

Parameter	150% target	200% target
Load resistance per phase	7.05 Ω	5.29 Ω
Theoretical current at nominal phase voltage	18.84 A	25.10 A
Displayed phase current RMS	18.12 A	23.84 A
Displayed phase voltage RMS	128.0 V	126.4 V
Loading Percent	144.4%	189.9%
Apparent Power Estimate	6,959 VA	9,037 VA
Overload flag	1 (active)	1 (active)
Overcurrent flag	0 (inactive)	1 (active)
Overtemperature flag	1 (active)	1 (active)
Condition Score	47.46	24.1

Table 5.3: Overload and overcurrent test results

As the resistance is reduced from 7.05 Ω to 5.29 Ω per phase, the displayed current rises from 18.12 A to 23.84 A and the apparent-power estimate rises from 6.959 kVA to 9.037 kVA. The overload flag is active in both cases; the overcurrent flag activates only in the 200% target case. The condition score falls from 47.46 to 24.1. The dashboard temperature output also rises from 123.2 to 191.2, but its physical unit and scaling require confirmation.

5.5 Overvoltage and Undervoltage Tests:

The source voltage is changed from the nominal 400 V line-to-line to 440 V for the overvoltage test and 360 V for the undervoltage test. The ideal secondary line voltage is calculated from the stated 400:230 V line-voltage ratio. The dashboard values below are phase RMS values, not directly measured secondary line-to-line values.

5.5.1 Overvoltage Test:

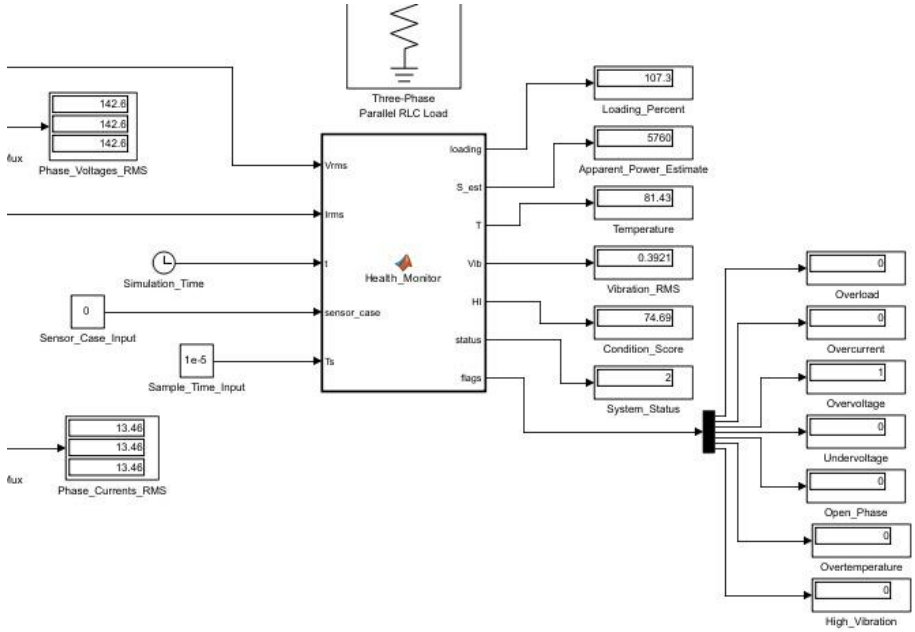


Figure 5.5.a: Monitoring outputs with 440 V source (overvoltage test)

5.5.2 Undervoltage Test:

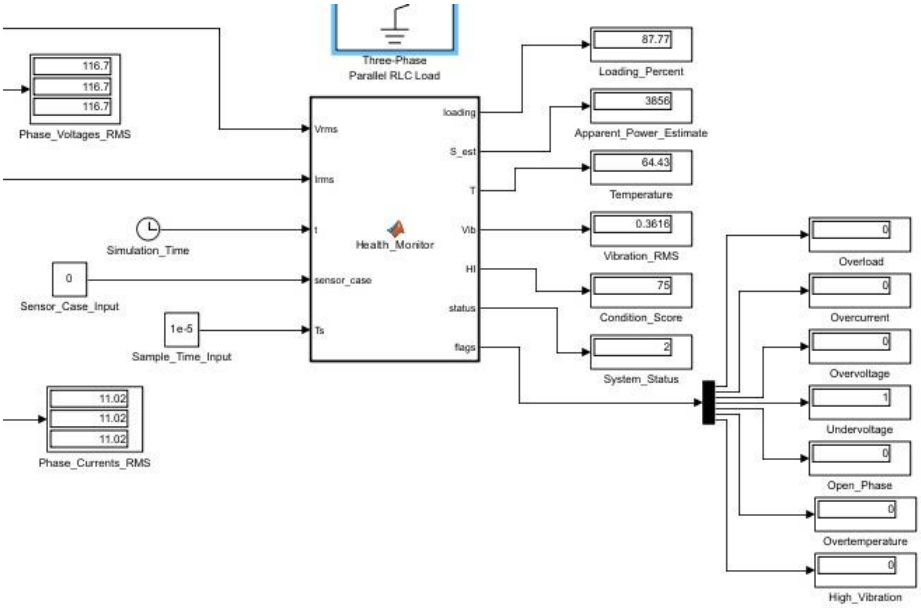


Figure 5.5.b: Monitoring outputs with 360 V source (undervoltage test)

Parameter	Undervoltage	Normal	Overvoltage
Primary/source line voltage	360 V	400 V	440 V
Ideal secondary line voltage	207 V	230 V	253 V
Ideal secondary phase voltage	119.52 V	132.79 V	146.07 V
Displayed phase RMS voltage	116.7 V	129.6 V	142.6 V
Voltage difference from ideal phase value	−2.36%	−2.40%	−2.38%
Voltage-related fault flag	Undervoltage = 1	Both = 0	Overvoltage = 1
Loading Percent	87.77%	97.53%	107.3%
Apparent power estimate	3,856 VA	4,760 VA	5,760 VA

Table 5.4: Voltage Variation Results

The displayed phase voltage follows the source variation: 116.7 V at 360 V input, 129.6 V at 400 V, and 142.6 V at 440 V. The corresponding undervoltage and overvoltage flags are active in the respective abnormal cases. The voltage values remain approximately 2.4% below the ideal phase-voltage references in all three cases.

5.6 Open-Phase Test :

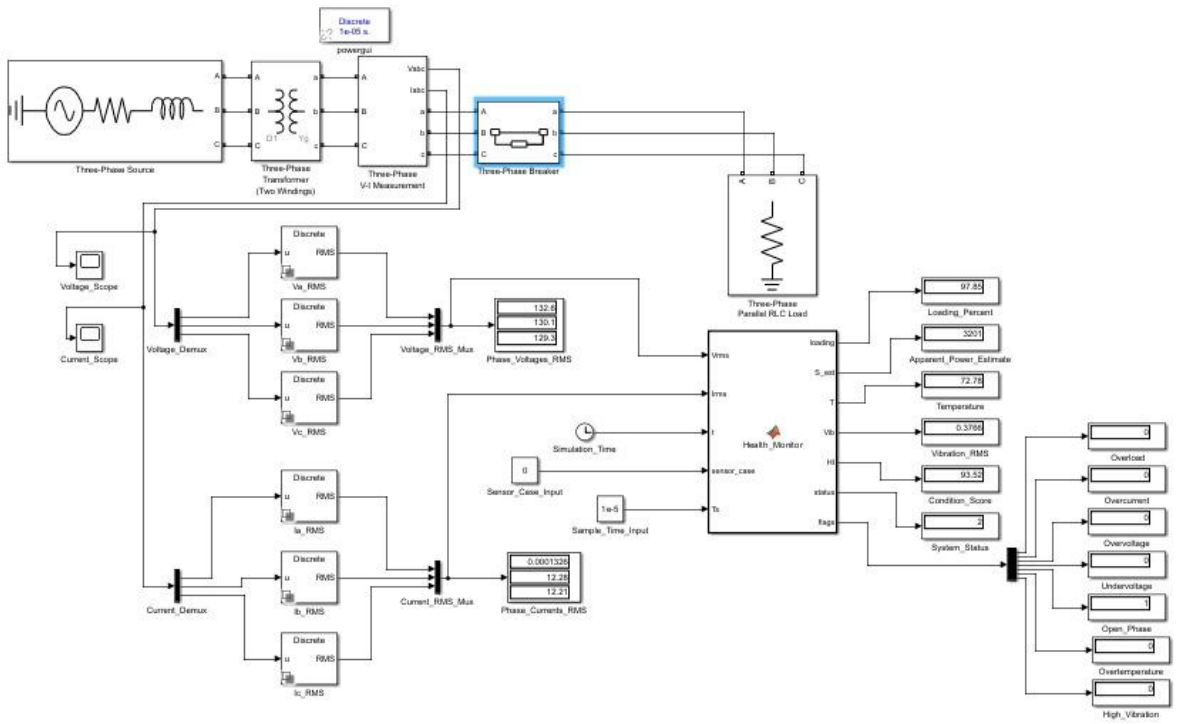


Figure 5.6: Model configuration with the breaker/fault location selected for the open-phase test

5.7 Short-Circuit Test :

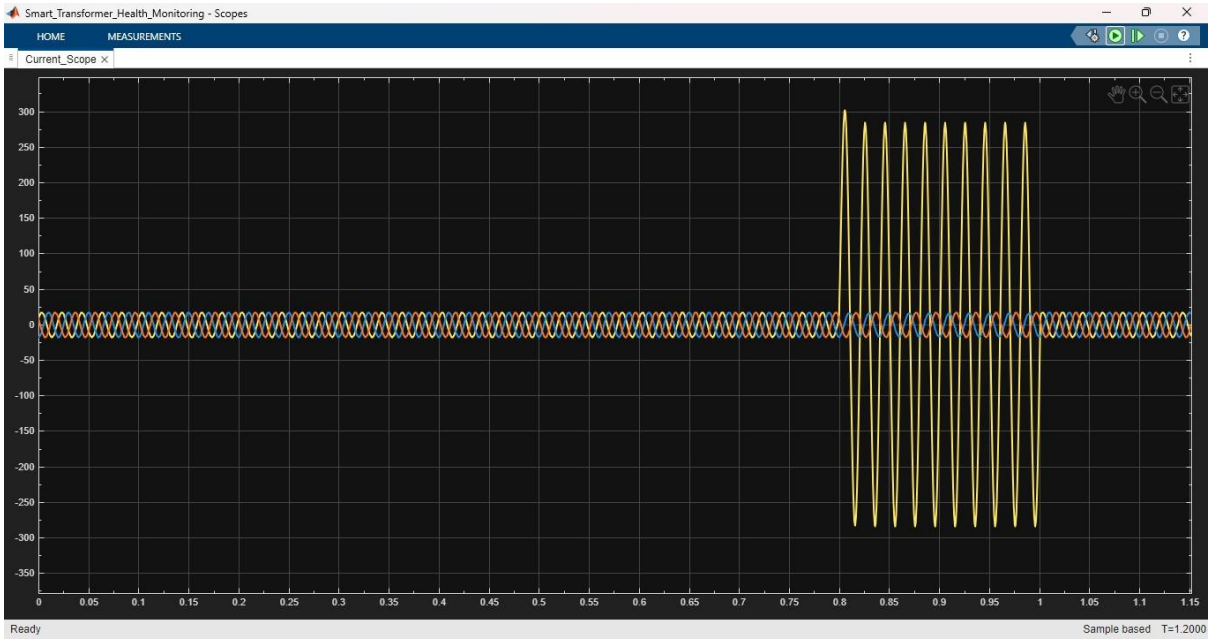


Figure 5.7: Current-scope response during the stated short-circuit interval (0.8–1.0 s)

5.8 Unbalanced-Load Test :

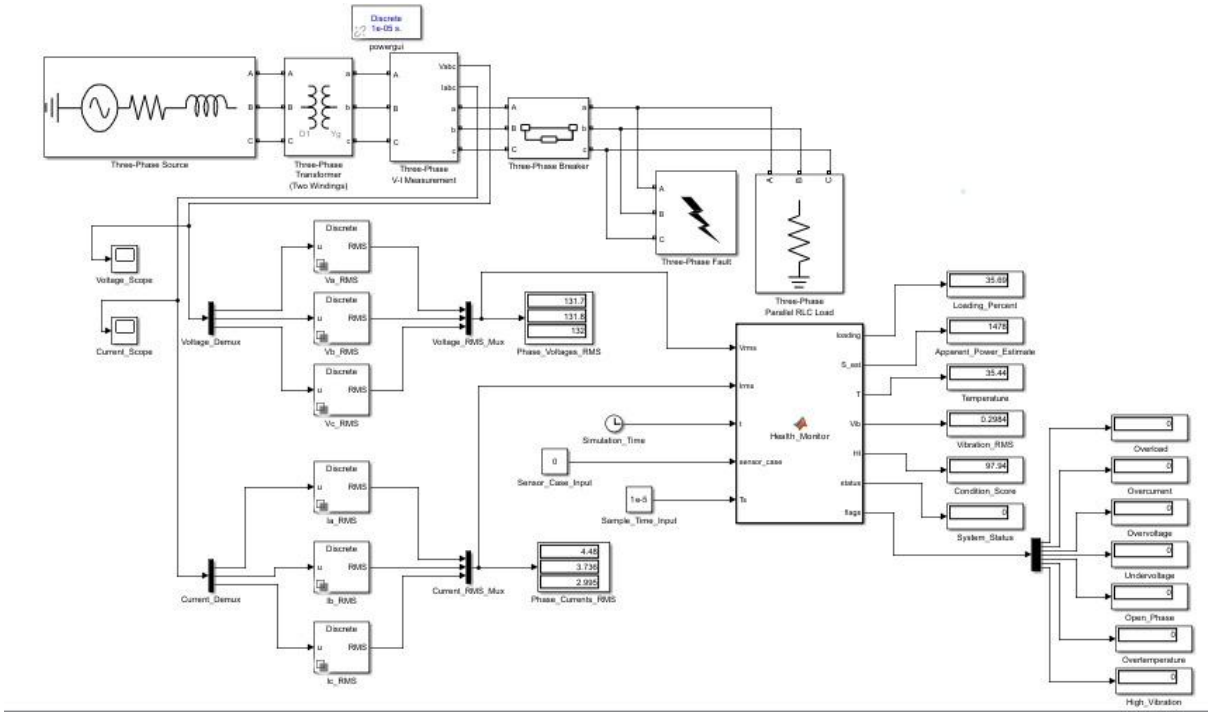


Figure 5.8.a: Simulation Response under an Unbalanced Load

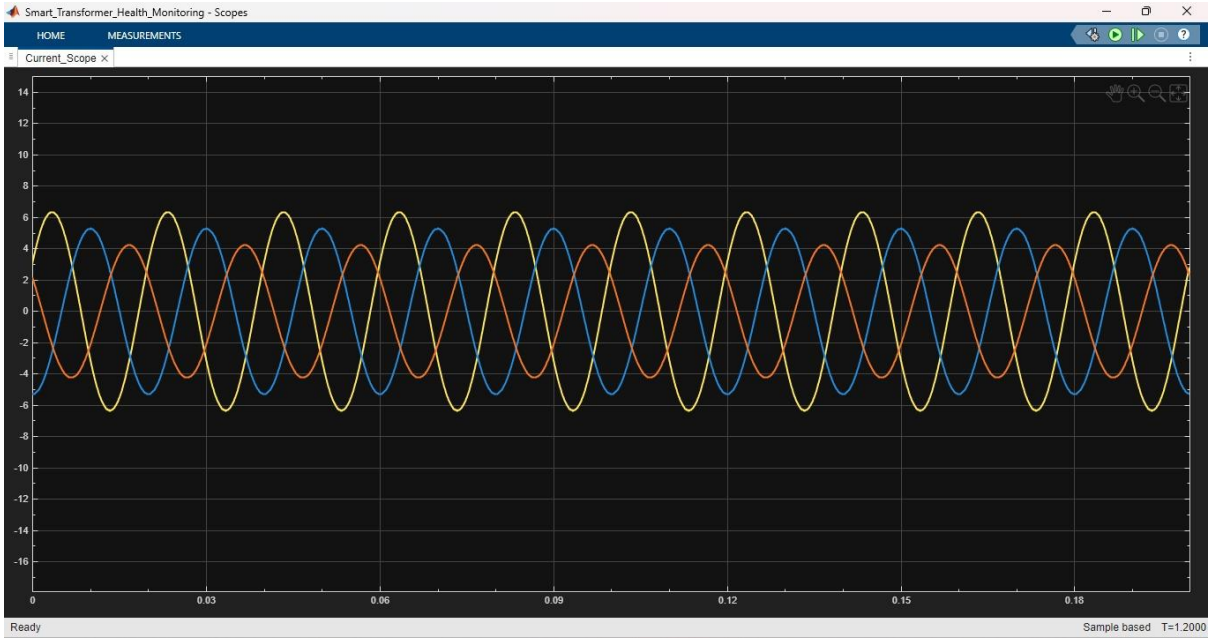


Figure 5.8.b: Three-phase current waveform under the unbalanced-load condition

5.9 Temperature Proxy Results :

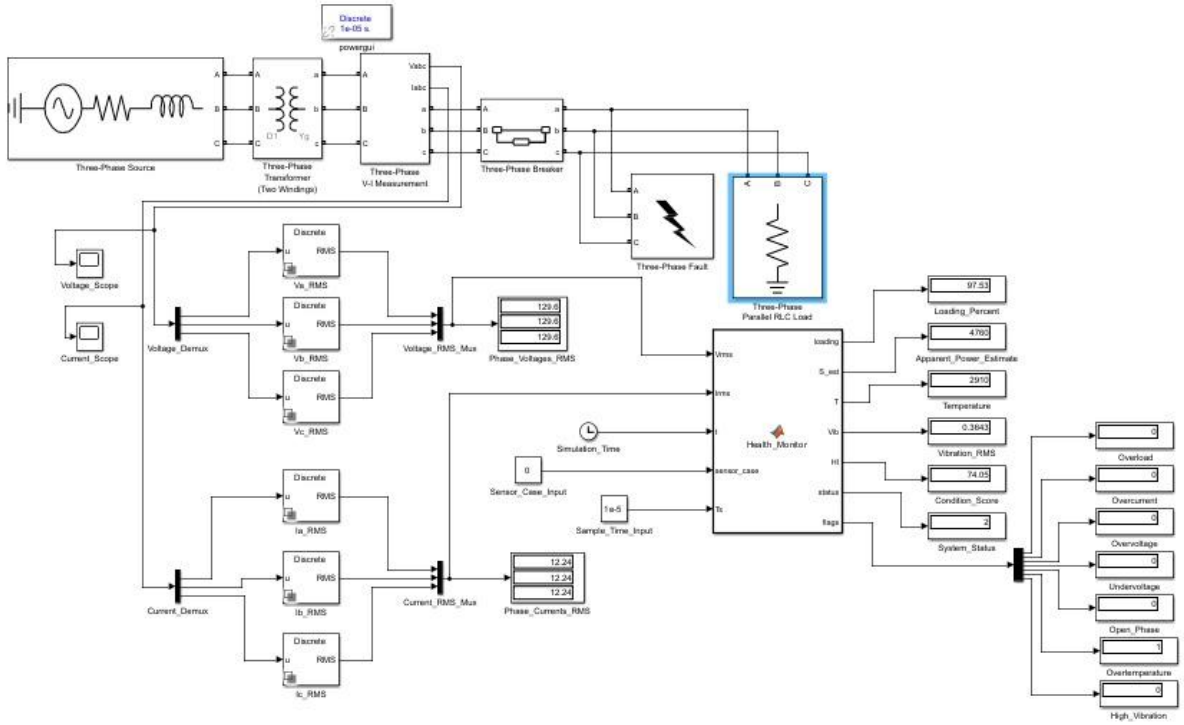


Figure 5.9: Estimated Temperature Response (temperature has gone upto 2910 degree C)

CHAPTER – 6

APPLICATIONS

6.1 Applications :

The proposed simulation-based transformer monitoring system has the following potential applications.

1. **Electrical Engineering Education:** Helps students understand transformer operation under normal, overload, and abnormal voltage conditions.
2. **Condition Monitoring:** Demonstrates how electrical measurements can be used to identify abnormal operating conditions.
3. **Prototype Development:** Provides a starting point for future implementation using sensors, a microcontroller, and a display.
4. **Performance Analysis:** Allows the effects of load changes and selected fault conditions to be studied.
5. **Future Monitoring Systems:** Provides a basic framework that could later be extended to other electrical equipment.

6.2 Advantages :

The proposed Smart Transformer Health Monitoring System offers the following advantages:

1. **Continuous Monitoring:** Enables monitoring of transformer parameters such as voltage and current through simulation.
2. **Early Fault Detection:** Helps identify abnormal operating conditions, such as overload, overvoltage, and undervoltage.
3. **Improved Reliability:** Helps understand how different operating conditions affect transformer performance.
4. **Reduced Maintenance Costs:** Early identification of abnormal conditions may help reduce maintenance costs in practical implementations.

5. **Real-Time Visualization:** MATLAB/Simulink scopes and monitoring displays can present measured parameters and simulation waveforms.
6. **Safe Testing Environment:** Allows selected fault conditions to be studied through simulation without deliberately creating electrical faults in physical equipment.
7. **Cost-Effective Development:** Provides a way to test monitoring concepts before investing in physical sensors and hardware.
8. **Future Expandability:** The simulation framework can be extended in future work to include physical sensors, microcontrollers, IoT-based monitoring, and advanced fault-detection techniques.

6.2 Practical Limitations :

The proposed system is a simulation-based educational project. Its performance will depend on the accuracy of the model, selected parameters, and diagnostic logic.

The main limitations are:

- The simulation does not directly measure the condition of a real transformer.
- The temperature model provides an estimate rather than a validated winding hot-spot temperature.
- The vibration signal is synthetic and does not represent actual transformer vibration.
- Assumed parameters may differ from those of a real transformer.
- Real-world implementation would require calibrated sensors, suitable signal conditioning, electrical isolation, and testing.
- The health indicator requires validation before it can be used for actual transformer condition assessment.

CHAPTER – 7

CONCLUSION AND FUTURE SCOPES

7.1 Conclusion :

This report defines a simulation-first engineering design for a Smart Transformer Health Monitoring System. The baseline combines a three-phase 5 kVA Δ–Y transformer model with voltage/current/power measurement, adjustable loading, controlled fault scenarios and a separate monitoring subsystem. The calculation sheet provides expected nominal quantities against which the teammate can validate the first simulation runs.

A key design principle is to distinguish measured electrical outputs from estimated or synthetic indicators. Electrical voltage, current and power can be obtained from Simscape sensors; temperature and vibration are represented by proxy models in this semester's scope. The health indicator and detection thresholds remain configurable educational logic and are not a substitute for asset-specific diagnostic practice.

The final project result will be established only when the actual model compiles, executes and produces the blank output evidence described in Appendix B. This draft intentionally does not claim that the MATLAB simulation has already been executed or validated.

7.2 Future scope :

- Use actual transformer nameplate and open-circuit/short-circuit test data to parameterize the transformer model.
- Add measurement noise, sensor accuracy limits, quantization, sampling and calibration effects.
- Compare multiple thermal models and validate parameters using measured temperature response.
- Replace synthetic vibration with accelerometer data and apply frequency-domain features when real data is available.
- Integrate MATLAB/Simulink output with an ESP32-based low-voltage hardware demonstrator in the next semester.
- Add event logging, data export, trend visualization and a carefully validated health-scoring method.
- Extend the architecture to multiple equipment assets only after the single-transformer baseline is validated.

REFERENCE

[1] International Electrotechnical Commission (IEC), *IEC Standard Publication*. Available: https://webstore.iec.ch/en/publication/588 (accessed 16 October 2026).

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APPENDIX A

DESIGN CALCULATIONS:

This appendix presents the theoretical calculations used to establish the operating parameters of the proposed three-phase transformer model.

A.1 Given parameters :

The following values are selected as the initial design parameters for the simulation.

Parameter	Value
Three-phase source voltage	400 V line-to-line RMS
Supply frequency	50 Hz
Source resistance, Rs	0.1 Ω per phase
Source inductance, Ls	1 μH per phase
Transformer rating	5 kVA
Rated voltage ratio	400/230 V line-to-line
Winding connection	Primary Δ; secondary grounded Y
Simulation sample time	1×10^{-5} s
Nominal load reference	10.58 Ω per phase

Table A.1: Transformer Design Specifications

A.2 Rated Line Current and Secondary Phase Voltages :

For a three-phase transformer, the apparent power is given by:

$$S = \sqrt{3}\, V_L I_L$$

---

1. Primary Rated Line Current

Given:

- Transformer rating, $S = 5000$ VA
- Primary line voltage, $V_{L1} = 400$ V

The primary rated line current is:

$$I_1 = \frac{S}{\sqrt{3}\, V_{L1}}$$

$$I_1 = \frac{5000}{\sqrt{3} \times 400}$$

$$\boxed{I_1 = 7.2169 \text{ A} \approx 7.22 \text{ A}}$$

2. Secondary Rated Line Current

Given:

- Transformer rating, $S = 5000$ VA
- Secondary line voltage, $V_{L2} = 230$ V

The secondary rated line current is:

$$I_2 = \frac{S}{\sqrt{3}\, V_{L2}}$$
$$I_2 = \frac{5000}{\sqrt{3} \times 230}$$
$$\boxed{I_2 = 12.5511 \text{ A} \approx 12.55 \text{ A}}$$

---

3. Secondary Phase-to-Neutral Voltage

For a star-connected secondary winding:

$$V_{ph} = \frac{V_L}{\sqrt{3}}$$
$$V_{2,ph} = \frac{230}{\sqrt{3}}$$
$$\boxed{V_{2,ph} = 132.7906 \text{ V} \approx 132.79 \text{ V}}$$

A.3 Source Impedance :

Given:

- Source resistance, $R_s = 0.1\ \Omega$
- Source inductance, $L_s = 1$ mH $= 0.001$ H
- Supply frequency, $f = 50$ Hz

1. Inductive Reactance

The inductive reactance is calculated as:

$$X_s = 2\pi f L_s$$

Substituting the given values:

$$X_s = 2\pi \times 50 \times 0.001$$
$$\boxed{X_s = 0.31416\ \Omega}$$

---

2. Magnitude of Source Impedance

For a series R-L circuit:

$$| Z_s | = \sqrt{R_s^2 + X_s^2}$$

Substituting the values:

$$| Z_s | = \sqrt{(0.1)^2 + (0.31416)^2}$$
$$| Z_s | = \sqrt{0.01 + 0.09870}$$
$$\boxed{| Z_s | \approx 0.3297\ \Omega \approx 0.330\ \Omega}$$

Therefore, the source impedance magnitude is approximately **0.330 Ω per phase** at 50 Hz. This calculation excludes the transformer's internal winding and leakage impedance.

A.4 Reference Load Resistance for 5 kW :

For a balanced, star-connected, purely resistive three-phase load:

$$P = 3V_{ph}^2 / R_{ph}$$

Rearranging the equation to calculate the phase resistance:

$$R_{ph} = \frac{3V_{ph}^2}{P}$$

Given:

- Total three-phase power, $P = 5000$ W
- Phase voltage, $V_{ph} = 132.7906$ V

Substituting the values:

$$R_{ph} = \frac{3 \times (132.7906)^2}{5000}$$
$$R_{ph} = \frac{3 \times 17633.94}{5000}$$
$$R_{ph} = \frac{52901.82}{5000}$$
$$\boxed{R_{ph} \approx 10.58\ \Omega}$$

Therefore, the theoretical resistance required for a balanced, star-connected resistive load consuming approximately 5 kW is **10.58 Ω per phase**, assuming 132.79 V across each phase resistance.

A.5 Overload Target Resistance and Theoretical Current :

For a purely resistive load operating at constant voltage, the resistance required for a target loading multiple $m$is:

$$R_m = \frac{R_{\text{rated}}}{m}$$

The theoretical current is calculated using Ohm's law:

$$I_m = \frac{V_{ph}}{R_m}$$

---

1. Calculation for 150% Overload -

Given:

- Rated resistance, $R_{\text{rated}} = 10.58\ \Omega$
- Overload factor, $m = 1.5$
- Phase voltage, $V_{ph} = 132.7906\ V$

Step 1: Calculate the load resistance -

$$R_{150\%} = \frac{R_{\text{rated}}}{1.5}$$
$$R_{150\%} = \frac{10.58}{1.5}$$
$$\boxed{R_{150\%} = 7.05\ \Omega}$$

Step 2: Calculate the theoretical current -

$$I_{150\%} = \frac{V_{ph}}{R_{150\%}}$$
$$I_{150\%} = \frac{132.7906}{7.05}$$
$$\boxed{I_{150\%} \approx 18.84\ A}$$

---

2. Calculation for 200% Overload -

Given:

- Rated resistance, $R_{\text{rated}} = 10.58\ \Omega$
- Overload factor, $m = 2$
- Phase voltage, $V_{ph} = 132.7906\ V$

Step 1: Calculate the load resistance -

$$R_{200\%} = \frac{R_{\text{rated}}}{2}$$
$$R_{200\%} = \frac{10.58}{2}$$
$$\boxed{R_{200\%} = 5.29\ \Omega}$$

Step 2: Calculate the theoretical current -

$$I_{200\%} = \frac{V_{ph}}{R_{200\%}}$$
$$I_{200\%} = \frac{132.7906}{5.29}$$
$$\boxed{I_{200\%} \approx 25.10\ A}$$

---

3. Comparison with Simulation Results -

The actual phase voltages displayed during the overload tests are lower than the nominal phase voltage of 132.79 V. Therefore, the theoretical currents are recalculated using the displayed voltages.

For the 150% overload case:

$$I = \frac{V_{ph}}{R_{150\%}}$$
$$I = \frac{128.0}{7.05}$$
$$\boxed{I \approx 18.16\ A}$$

The displayed simulation current is **18.12 A**.

For the 200% overload case:

$$I = \frac{V_{ph}}{R_{200\%}}$$
$$I = \frac{126.4}{5.29}$$
$$\boxed{I \approx 23.89\ A}$$

The displayed simulation current is **23.84 A**.

A.6 Ideal secondary voltage under source variation :

Using the transformer voltage ratio of 400:230 V:

$$V_{2,LL} = V_{1,LL} \times \frac{230}{400}$$

For a star-connected secondary:

$$V_{2,ph} = \frac{V_{2,LL}}{\sqrt{3}}$$

Source voltage	Ideal secondary line voltage	Ideal secondary phase voltage	Displayed phase RMS voltage
360 V	$360 \times 230/400 = 207$ V	$207/\sqrt{3} = 119.51$ V	116.7 V
400 V	$400 \times 230/400 = 230$ V	$230/\sqrt{3} = 132.79$ V	129.6 V
440 V	$440 \times 230/400 = 253$ V	$253/\sqrt{3} = 146.07$ V	142.6 V

Table A.2: Theoretical and Displayed Secondary Voltage Comparison at Different Source Voltages

Voltage Deviation -

The percentage voltage deviation is:

$$\%\ \text{Deviation} = \frac{V_{\text{measured}} - V_{\text{ideal}}}{V_{\text{ideal}}} \times 100$$

For 360 V source:

$$\frac{116.7 - 119.51}{119.51} \times 100 = \boxed{-2.35\%}$$

For 400 V source:

$$\frac{129.6 - 132.79}{132.79} \times 100 = \boxed{-2.40\%}$$

For 440 V source:

$$\frac{142.6 - 146.07}{146.07} \times 100 = \boxed{-2.38\%}$$

The displayed secondary phase voltages are approximately 2.4% below their ideal calculated values in all three cases.

A.7 Cross-check of apparent-power estimate :

For a balanced three-phase system, apparent power is cross-checked using:

$$S \approx 3V_{ph}I_{ph}$$

The calculation uses the displayed phase RMS values and is not a separate measured output.

Operating case	Calculation	Calculated S	Dashboard estimate
Normal	3 × 129.6 × 12.24	4,760.9 VA	4,760 VA
150% target	3 × 128.0 × 18.12	6,958.1 VA	6,959 VA
200% target	3 × 126.4 × 23.84	9,037.4 VA	9,037 VA
Overvoltage	3 × 142.6 × 13.46	5,757.8 VA	5,760 VA
Undervoltage	3 × 116.7 × 11.02	3,856.5 VA	3,856 VA

Table A.3: Cross-Verification of Calculated and Dashboard Apparent-Power Estimates Under Different Operating Conditions

The calculated values closely agree with the dashboard estimates, with minor differences due to rounding of the displayed RMS values. The output is referred to as an **apparent-power estimate**, not active power.

A.8 Current-based loading cross-check :

Using the rated secondary line current, $I_{2,\text{rated}} = 12.5511$ A:

$$Loading\ (\%) = \frac{I_{measured}}{I_{2,rated}} \times 100$$

Case	Displayed current	Calculated loading	Dashboard loading
Normal	12.24 A	97.52%	97.53%
150% target	18.12 A	144.37%	144.4%
200% target	23.84 A	189.95%	189.9%
Overvoltage	13.46 A	107.24%	107.3%
Undervoltage	11.02 A	87.80%	87.77%

Table A.4: Cross-Verification of Calculated and Dashboard Loading Percentages

The calculated loading percentages closely match the dashboard values, with minor differences due to rounding. This confirms that loading is likely calculated from RMS current relative to the rated secondary current.

APPENDIX B

MODEL VALIDATION CHECKLIST :

Sr. No.	Validation Item	Status
1	Model opens in the selected MATLAB release.	Pass
2	Required electrical connections are complete.	Pass
3	Transformer voltage ratings and winding connections are configured correctly.	Pass
4	Normal secondary voltage is close to its theoretical value.	Pass
5	Measured current and power are reasonably consistent with the calculated references.	Pass
6	Measurement blocks are connected correctly.	Pass
7	Normal operating waveforms have been recorded.	Pass
8	Implemented overload tests produce an identifiable response.	Pass
9	Implemented voltage variation tests produce an identifiable response.	Pass
10	Additional implemented fault tests produce identifiable responses.	Pass
11	The simulation results and model settings have been documented.	Pass

Table B.1: Model Validation Checklist

----------End Of Report----------